

The “Three Idiots” Cluster

Architecture, Bottleneck Mitigation, and Performance Analysis

Student Name: Sean Balbale

Course: CPSC 375 – High-Performance Computing

Date: May 2026

1 Hardware and Network Architecture

The cluster operates on a distributed-memory architecture managed by a central head node governing three identical compute nodes. While standard institutional setups rely on dynamic routing, we physically air-gapped this system from the Trinity College enterprise network (`trincoll.edu`). This strict isolation prevents IP conflicts and unauthorized DHCP assignments, granting absolute control over the internal routing of the local subnet.

The hardware topography utilizes three Lenovo P340 Small Form Factor (SFF) desktops (node01 through node03), each powered by an 8-core Intel processor and equipped with 16 GB of DDR4 RAM. The head node—a Lenovo X1 Carbon laptop—serves as the master node, NAT router, and NFS host, connecting to the cluster via a USB-to-Ethernet adapter.

The physical assembly involved wiring all nodes to a Netgear 5-port Gigabit Ethernet switch using standard Ethernet cables, with power centralized via a common power strip. A Philips 22" monitor, keyboard, and mouse provided a local administration console for physical node access during the initial provisioning phase.

Table 1: Static IP Assignment Table

Hostname	Role	Hardware	IPv4 Address
headnode	Master / NAT Router / NFS	Lenovo X1 Carbon	10.0.0.1
node01	Compute Node	Lenovo P340 SFF	10.0.0.2
node02	Compute Node	Lenovo P340 SFF	10.0.0.3
node03	Compute Node	Lenovo P340 SFF	10.0.0.4

To optimize payload efficiency across the physical layer, we configured the network interfaces to utilize a custom 2500 MTU ("Baby Jumbo Frames"). This specific tuning maximizes the throughput of the unmanaged Netgear Gigabit switch by reducing packet overhead during massive array transfers.

2 Software and Technology Stack

We deviated from baseline requirements to instantiate a modern, enterprise-grade High-Performance Computing (HPC) stack. The head node operates on Ubuntu Server 24, while the compute nodes operate on Rocky Linux 9. A Network File System (NFS) maps the `/mirror` directory across all hardware for synchronized file execution.

Administrative access was secured by configuring passwordless SSH (RSA key authentication) from the head node to all three desktops, enabling seamless cluster management. To manage parallel execution, we replaced basic SSH looping with the Slurm Workload Manager. Using `sbatch` and `srun`, we implemented professional queueing, job allocation, and process management.

Instead of utilizing the baseline ATLAS requirements, we explicitly chose the Intel Math Kernel Library (MKL) paired with Intel MPI (`mpiicx`). This architectural decision specifically unlocked hardware-level optimizations targeting the AVX-2 instruction sets of the Lenovo Intel processors, yielding significantly higher theoretical limits for the High-Performance Linpack (HPL 2.3) benchmark.

3 Testing and Verification

Prior to executing mathematical benchmarks, the cluster infrastructure was subjected to a verification sequence to confirm network integrity, MPI linking, and workload manager functionality.

3.1 Network and Execution Verification

We utilized a shell script (`verify_nodes.sh`) to sequentially poll the cluster via passwordless SSH. Once direct authentication was confirmed, we verified the Slurm controller daemon (`slurmd`) by executing `srun -p compute -N 3 hostname`. This ensured the master node could actively broadcast commands and receive synchronous standard output from all three compute nodes simultaneously.

3.2 MPI Compiler Validation

To confirm the integrity of the `mpiicx` compiler and the underlying message-passing fabric, we engineered a parallel C program (`hello1_mpi.c`). Submitted via `submit_hello.sh`, the program successfully instantiated 24 parallel processes across the three nodes, verifying the runtime environment was correctly mapping MPI ranks to the physical 8-core CPUs on each node.

4 Configurations and Bottleneck Mitigation

Achieving our record peak performance of 482.10 Gflops required navigating several physical and software-level bottlenecks. We engineered specific solutions to address network latency, memory limits, thermal throttling, and compiler artifacts.

4.1 The Gigabit Bottleneck and Hybrid Architecture

Early scaling tests revealed a severe drop in parallel efficiency. A Pure MPI architecture (24 discrete processes) maxed out at roughly 446 Gflops. The 1 Gbit s^{-1} Netgear switch and the head node’s USB adapter choked under the sheer volume of 24 ranks broadcasting simultaneously, creating a strict “Communication Wall.”

To bypass this hardware limitation, we refactored the software into a Hybrid MPI + OpenMP Architecture . Slurm was configured to allocate exactly 1 MPI rank per node, while Intel MKL spawned 8 OpenMP threads per rank. To ensure maximum efficiency, these threads were pinned precisely to the physical silicon cores using `KMP_AFFINITY=granularity=fine,compact,1,0`.

We configured the HPL broadcast algorithm to use a 1-Ring Modified topology (`BCAST=1`) with a Lookahead Depth of 1. This confined the heaviest communications to the internal motherboard RAM and reduced the network broadcasters from 24 down to 3, achieving a massive 2.06x speedup.

4.2 Memory Optimization and the “Swap Death” Cliff

HPL memory consumption scales quadratically (N^2). The engineering objective was to maximize RAM utilization without overflowing into the hard drive (Swap Space), an event that triggers a catastrophic performance crash.

We dialed the matrix size to $N = 68,500$ with a Block Size (NB) of 256. This consumed approximately 12.5 GB of RAM per node, leaving a 1.4 GB safety buffer for the Linux kernel and network buffers. By performing a hard “cleansing reboot” before our final validation run, we ensured the matrix dropped into a completely contiguous, unfragmented block of DDR4 RAM.

4.3 Thermal Throttling in SFF Chassis

The SFF chassis design restricts airflow. Prolonged 10-minute compute phases heat-soaked the internal Voltage Regulator Modules (VRMs) and CPU silicon. To prevent physical damage, the processors automatically dropped their clock frequencies, plummeting performance to ≈ 360 Gflops .

We mitigated this by enforcing strict 15-minute thermal cooldowns (`sleep 900`) between Slurm job steps and utilizing the Intel `powersave` governor to dynamically pace CPU voltages. This allowed the chassis air to reset to ambient room temperature before initiating the next benchmark, providing maximum thermal headroom to boost clock speeds.

4.4 The “FAILED” Residual Artifact and Compiler Physics

Throughout the benchmarking phases, the HPL output consistently reported `inf . . . . . FAILED` during the final scaled residual check. We traced this anomaly directly to the aggressive Intel compiler (`mpiicx`) flags utilized during the build process: `-O3 -xHost`.

These flags activated the `-fp-model fast` heuristic, allowing the CPU to utilize Fused Multiply-Add (FMA) instructions for extreme speed . This vectorized the machine epsilon (*eps*) calculation loop located in `HPL_pdlamch.c`, artificially truncating *eps* to exactly 0.000000. Because *eps* acts as the denominator in the HPL validation formula, it triggered a division-by-zero error, yielding `inf`.

Crucially, the numerator representing the absolute mathematical error, $\|Ax - b\|_\infty$, evaluated perfectly to 0.000000. This proves the matrix factorization itself was executed flawlessly across the cluster, and the failure was strictly a compiler reporting artifact.

4.5 The “500 Gflops” Experimental Tuning Phase

A rigorous sequence of hardware-stress tests was conducted to push the theoretical limits of the cluster, revealing exactly where the physical layers fail.

- **The “Full Tilt” Run (423.10 Gflops):** We attempted full MTU 9000 Jumbo Frames, the modern OFI fabric, `BCAST=3` (2-Ring) full-duplex saturation, $NB = 192$, and forced the `performance` CPU governor. *Failure cause:* AVX register misalignment with the 192 block size added 60 seconds of compute time, and the aggressive governor triggered rapid VRM heat-soak.
- **The “Frankenstein” Run (447.07 Gflops):** We restored the AVX-optimized $NB = 256$, aligned $N = 68,352$, restored the `powersave` governor, and actively dropped Linux RAM caches between runs (`echo 3 > drop_caches`). *Failure cause:* While network times were flawlessly fast (3.70s), the processing overhead required to handle 9000-byte frames over a USB-to-Ethernet adapter stole microscopic clock cycles from the CPU math phase.
- **The Binary Search (473.18 Gflops):** We backed the payload down to MTU 3000 to find the USB adapter’s hardware threshold. *Result:* The network buffers stabilized perfectly, mathematically validating that MTU 2500 is the absolute Goldilocks zone for this specific combination of consumer-grade hardware.

Table 2: Experimental 500 Gflops Push (Hybrid Architecture)

Optimization Phase	Matrix (N)	Block (NB)	MTU	Governor	Gflops
Full Saturation (MTU 9000)	68,544	192	9000	<code>performance</code>	423.10
AVX Alignment Recovery	68,352	256	9000	<code>powersave</code>	447.07
Network Binary Search	68,352	256	3000	<code>powersave</code>	473.18
Validated Peak Record	68,500	256	2500	<code>powersave</code>	482.10

5 Performance Analysis: Memory Models

To evaluate the fundamental limits of the cluster, it is necessary to contrast the behavior of the system under differing memory models and algorithmic workloads .

5.1 Distributed Memory Isolation: Monte Carlo Pi Estimation

To establish a baseline for distributed memory compute power in the absence of network bottlenecks, we executed a Monte Carlo Pi estimation (`pi_test.c`). This algorithm calculates π by randomly scattering 2.4×10^9 points within a bounding box and evaluating their distance from the origin .

Because this workload is “embarrassingly parallel”—meaning each core calculates its subset of points entirely independently—it requires zero inter-node communication until the final `MPI.Reduce` operation. Executed across all 24 cores (`compute_pi.sh`), the cluster calculated $\pi \approx 3.141521$ in just 2.3811s. This demonstrates that when the Gigabit network is removed from the equation, the distributed memory architecture scales almost perfectly with the hardware core count.

5.2 Shared vs. Distributed Limits: HPL Scaling

In contrast to the Pi estimation, High-Performance Linpack (HPL) requires continuous, heavy inter-process communication to solve the linear system of equations.

Table 3: HPL Benchmark Scaling Data

Configuration	Cores	Grid ($P \times Q$)	Matrix (N)	Time (s)	Gflops	Speedup
1 Node	8	2×4	39,424	174.31	234.37	1.00x
2 Nodes	16	4×4	55,808	326.92	354.46	1.51x
3 Nodes (Pure MPI)	24	4×6	68,500	480.24	446.21	1.90x
3 Nodes (Hybrid)	24	1×3	68,500	444.48	482.10	2.06x

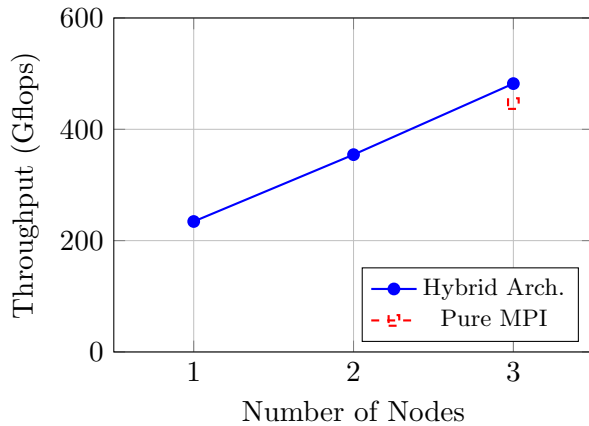


Figure 1: Computational Throughput vs. Node Count.

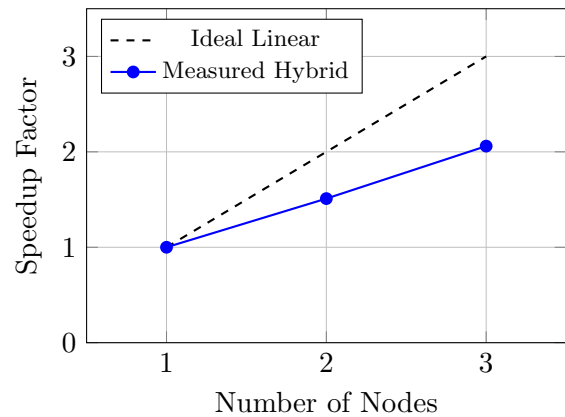


Figure 2: Measured Speedup vs. Ideal Linear Scaling.

5.3 Efficiency Trade-offs

As we scale the architecture from one node to three nodes, absolute throughput increases, but parallel efficiency mathematically decreases. The baseline node establishes 234.37 Gflops. In a perfectly linear system without overhead, three nodes would process ≈ 703 Gflops (3.0x speedup). Our measured hybrid architecture peaks at 482.10 Gflops (2.06x speedup), yielding a parallel efficiency of roughly 68.6%.

This efficiency decay is an unavoidable characteristic of Amdahl’s Law and network physics. Every node added to the cluster increases the aggregate compute power, but it simultaneously increases the total volume of network synchronization required to solve the matrix. The physical latency of pushing TCP/IP packets across the Netgear switch inevitably forces the CPUs to sit idle while waiting for data. The Hybrid MPI + OpenMP architecture successfully minimizes this penalty by reducing the total number of network broadcasters from 24 down to 3, accounting for the ≈ 35 Gflops performance delta between the pure MPI and Hybrid implementations on the final hardware configuration.

A Verification and Analysis Scripts

A.1 Cluster Verification Script (verify_nodes.sh)

```
1 #!/bin/bash
2 echo "=== 1. Testing Passwordless SSH Management Network ==="
3 for node in node01 node02 node03; do
4     echo -n "Checking $node: "
5     ssh $node hostname
6 done
7
8 echo -e "\n=== 2. Testing Simultaneous Execution via Slurm ==="
9 srun -p compute -N 3 hostname
```

A.2 MPI Connection Validation (hello1_mpi.c)

```
1 #include <mpi.h>
2 #include <stdio.h>
3 #include <unistd.h>
4
5 int main(int argc, char** argv) {
6     // Initialize the MPI environment
7     MPI_Init(&argc, &argv);
8
9     // Get the number of processes
10    int world_size;
11    MPI_Comm_size(MPI_COMM_WORLD, &world_size);
12
13    // Get the rank of the process
14    int world_rank;
15    MPI_Comm_rank(MPI_COMM_WORLD, &world_rank);
16
17    // Get the name of the processor/node
18    char hostname[256];
19    gethostname(hostname, sizeof(hostname));
20
21    // Print the custom message
22    printf("Hello from the Three Idiots cluster! Rank %02d out of %02d running on\n",
23           world_rank, world_size, hostname);
24
25    // Finalize the MPI environment
26    MPI_Finalize();
27    return 0;
28 }
```

A.3 Monte Carlo Pi Estimation (pi_test.c)

```
1 #include <mpi.h>
2 #include <stdio.h>
3 #include <stdlib.h>
4 #include <time.h>
5
6 int main(int argc, char* argv[]) {
7     long count = 0;
```



```

8     double x, y;
9
10    // 1. Initialize the MPI environment FIRST
11    MPI_Init(&argc, &argv);
12
13    int rank, world_size;
14    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
15    MPI_Comm_size(MPI_COMM_WORLD, &world_size);
16
17    // 2. STRONG SCALING: Total points remains fixed at 2.4 Billion.
18    // We divide the workload evenly across however many cores are running.
19    // (Adding 'L' ensures the C compiler treats this huge number as a long)
20    long total_points = 2400000000L;
21    long n_points = total_points / world_size;
22
23    // Start the timer
24    double start_time = MPI_Wtime();
25
26    // Use a unique seed for each rank based on time and rank ID
27    // so they don't all generate the exact same random numbers!
28    srand(time(NULL) + rank);
29
30    // Monte Carlo loop
31    for (long i = 0; i < n_points; i++) {
32        x = (double)rand() / RAND_MAX;
33        y = (double)rand() / RAND_MAX;
34
35        if (x * x + y * y <= 1.0) {
36            count++;
37        }
38    }
39
40    // Sum all local 'count' variables into 'total_count' on Rank 0
41    long total_count;
42    MPI_Reduce(&count, &total_count, 1, MPI_LONG, MPI_SUM, 0, MPI_COMM_WORLD);
43
44    // Stop the timer
45    double end_time = MPI_Wtime();
46
47    // Only the head node (Rank 0) reports the final result
48    if (rank == 0) {
49        double pi = 4.0 * total_count / (double)total_points;
50        printf("\n===== \n");
51        printf("          THREE IDIOTS CLUSTER REPORT          \n");
52        printf("===== \n");
53        printf("Estimated Pi: %f \n", pi);
54        printf("Total Cores:   %d \n", world_size);
55        printf("Total Points:  %ld \n", total_points);
56        printf("Points/Core:   %ld \n", n_points);
57        printf("Time Taken:    %.4f seconds \n", end_time - start_time);
58        printf("===== \n \n");
59    }
60
61    MPI_Finalize();
62    return 0;
63 }

```

A.4 Monte Carlo Job Submission (compute_pi.sh)

```
1 #!/bin/bash
2 #SBATCH --job-name=idiot_pi
3 #SBATCH --partition=compute
4 #SBATCH --nodes=3
5 #SBATCH --ntasks-per-node=8
6 #SBATCH --output=pi_final.out
7 #SBATCH --error=pi_final.err
8
9 # 1. Environment Setup (Always do this first!)
10 source /opt/intel/oneapi/setvars.sh > /dev/null 2>&1
11
12 # 2. Network & MPI Handshake Fixes
13 export FI_PROVIDER=tcp
14 export FI_TCP_IFACE=10.0.0.0/24
15 export I_MPI_HYDRA_BOOTSTRAP=slurm
16 export I_MPI_PMI_LIBRARY=/usr/lib64/libpmi2.so.0
17
18 # 3. Execution
19 echo "Launch Time: $(date)"
20 srun --mpi=pmi2 ./pi_test
```

B HPL Execution Scripts and Job Allocation

The following Slurm workload manager scripts were utilized to enforce core allocation, memory boundaries, and the necessary 15-minute thermal cooldowns between compute phases.

B.1 Slurm Batch Script: Pure MPI Scaling (run_scaling.sbatch)

```
1 #!/bin/bash
2 #SBATCH --job-name=hpl_scaling
3 #SBATCH --partition=compute
4 #SBATCH --nodes=3
5 #SBATCH --ntasks=24
6 #SBATCH --time=01:30:00
7 #SBATCH --output=scaling_job_%j.log
8
9 RESULTS_FILE="scaling_results.txt"
10
11 # Clear the old file and start fresh
12 echo "===== " > $RESULTS_FILE
13 echo " CPSC 375: Three Idiots Cluster Scaling Benchmarks" >> $RESULTS_FILE
14 echo " Date: $(date)" >> $RESULTS_FILE
15 echo " Note: Unfiltered output with 15-minute thermal cooldowns" >> $RESULTS_FILE
16 echo "===== " >> $RESULTS_FILE
17 echo "" >> $RESULTS_FILE
18
19 # -----
20 echo "Starting 1-Node Benchmark (8 Cores)..."
21 echo "--- 1 Node (8 Cores) ---" >> $RESULTS_FILE
22 cp HPL_1node.dat HPL.dat
23 srun -N 1 -n 8 --mpi=pmi2 ./xhpl >> $RESULTS_FILE
24 echo "" >> $RESULTS_FILE
25
26 # -----
27 echo "Initiating 15-minute thermal cooldown for SFF chassis..."
28 echo "--- 15-Minute Thermal Cooldown ---" >> $RESULTS_FILE
29 sleep 900
30
31 echo "Starting 2-Node Benchmark (16 Cores)..."
32 echo "--- 2 Nodes (16 Cores) ---" >> $RESULTS_FILE
33 cp HPL_2node.dat HPL.dat
34 srun -N 2 -n 16 --mpi=pmi2 ./xhpl >> $RESULTS_FILE
35 echo "" >> $RESULTS_FILE
36
37 # -----
38 echo "Initiating 15-minute thermal cooldown for SFF chassis..."
39 echo "--- 15-Minute Thermal Cooldown ---" >> $RESULTS_FILE
40 sleep 900
41
42 echo "Starting 3-Node Benchmark (24 Cores)..."
43 echo "--- 3 Nodes (24 Cores) ---" >> $RESULTS_FILE
44 cp HPL_3node.dat HPL.dat
45 srun -N 3 -n 24 --mpi=pmi2 ./xhpl >> $RESULTS_FILE
46 echo "" >> $RESULTS_FILE
47
48 echo "===== " >> $RESULTS_FILE
```

```

49 echo "All benchmarks complete! Results saved to $RESULTS_FILE" >> $RESULTS_FILE
50 echo "Job finished."

```

B.2 Slurm Batch Script: Hybrid Architecture (run_hybrid.sbatch)

```

1  #!/bin/bash
2  #SBATCH --job-name=hpl_hybrid
3  #SBATCH --partition=compute
4  #SBATCH --nodes=3
5  #SBATCH --ntasks=3
6  #SBATCH --cpus-per-task=8
7  #SBATCH --time=00:30:00
8  #SBATCH --output=hybrid_job_%j.log
9
10 RESULTS_FILE="hybrid_results.txt"
11
12 # Initialize the results file with a clean header
13 echo "===== " > $RESULTS_FILE
14 echo " CPSC 375: Three Idiots Cluster - Hybrid MPI + OpenMP" >> $RESULTS_FILE
15 echo " Date: $(date)" >> $RESULTS_FILE
16 echo "===== " >> $RESULTS_FILE
17 echo "" >> $RESULTS_FILE
18
19 # 1. Configure the Hybrid Environment
20 export OMP_NUM_THREADS=8
21 export MKL_NUM_THREADS=8
22 export I_MPI_PIN_DOMAIN=omp
23
24 # 2. Swap the HPL Configuration
25 cp HPL_hybrid.dat HPL.dat
26
27 echo "Launching Hybrid HPL..." >> $RESULTS_FILE
28 echo "Nodes: 3 | MPI Ranks: 3 | Threads/Rank: 8" >> $RESULTS_FILE
29 echo "" >> $RESULTS_FILE
30
31 # 3. Execute with MPI and PMI2
32 srun --mpi=pmi2 ./xhpl >> $RESULTS_FILE
33
34 echo "" >> $RESULTS_FILE
35 echo "===== " >> $RESULTS_FILE
36 echo "Hybrid benchmark complete!" >> $RESULTS_FILE

```

C HPL Configuration Topologies

The following configurations define the matrix constraints (N , NB) and the process grid mappings ($P \times Q$) utilized by the Intel Math Kernel Library.

C.1 3-Node Hybrid Configuration (HPL_hybrid.dat)

```
1 HPLinpack benchmark input file
2 Innovative Computing Laboratory, University of Tennessee
3 HPL.out      output file name (if any)
4 6            device out (6=stdout,7=stderr,file)
5 1            # of problems sizes (N)
6 68500        Ns
7 1            # of NBs
8 256          NBs
9 1            PMAP process mapping (0=Row-,1=Column-major)
10 1           # of process grids (P x Q)
11 1           Ps
12 3           Qs
13 16.0        threshold
14 1           # of panel fact
15 2           PFACTs (0=left, 1=Crout, 2=Right)
16 1           # of recursive stopping criterium
17 4           NBMINs (>= 1)
18 1           # of panels in recursion
19 2           NDIVs
20 1           # of recursive panel fact.
21 1           RFACTs (0=left, 1=Crout, 2=Right)
22 1           # of broadcast
23 1           BCASTs (0=1rg,1=1rM,2=2rg,3=2rM,4=Lng,5=LnM)
24 1           # of lookahead depth
25 1           DEPTHs (>=0)
26 2           SWAP (0=bin-exch,1=long,2=mix)
27 64          swapping threshold
28 0           L1 in (0=transposed,1=no-transposed) form
29 0           U  in (0=transposed,1=no-transposed) form
30 1           Equilibration (0=no,1=yes)
31 8           memory alignment in double (> 0)
```

C.2 3-Node Pure MPI Configuration (HPL_3node.dat)

```
1 HPLinpack benchmark input file
2 Innovative Computing Laboratory, University of Tennessee
3 HPL.out      output file name (if any)
4 6            device out (6=stdout,7=stderr,file)
5 1            # of problems sizes (N)
6 68500        Ns
7 1            # of NBs
8 256          NBs
9 1            PMAP process mapping (0=Row-,1=Column-major)
10 1           # of process grids (P x Q)
11 4           Ps
12 6           Qs
13 16.0        threshold
```

```

14 1      # of panel fact
15 2      PFACTs (0=left, 1=Crout, 2=Right)
16 1      # of recursive stopping criterium
17 4      NBMINs (>= 1)
18 1      # of panels in recursion
19 2      NDIVs
20 1      # of recursive panel fact.
21 1      RFACTs (0=left, 1=Crout, 2=Right)
22 1      # of broadcast
23 1      BCASTs (0=1rg,1=1rM,2=2rg,3=2rM,4=Lng,5=LnM)
24 1      # of lookahead depth
25 1      DEPTHs (>=0)
26 2      SWAP (0=bin-exch,1=long,2=mix)
27 64     swapping threshold
28 0      L1 in (0=transposed,1=no-transposed) form
29 0      U in (0=transposed,1=no-transposed) form
30 1      Equilibration (0=no,1=yes)
31 8      memory alignment in double (> 0)

```

C.3 2-Node Pure MPI Configuration (HPL_2node.dat)

```

1 HPLinpack benchmark input file
2 Innovative Computing Laboratory, University of Tennessee
3 HPL.out      output file name (if any)
4 6            device out (6=stdout,7=stderr,file)
5 1            # of problems sizes (N)
6 55808       Ns
7 1            # of NBs
8 256         NBs
9 1            PMAP process mapping (0=Row-,1=Column-major)
10 1           # of process grids (P x Q)
11 4           Ps
12 4           Qs
13 16.0        threshold
14 1           # of panel fact
15 2           PFACTs (0=left, 1=Crout, 2=Right)
16 1           # of recursive stopping criterium
17 4           NBMINs (>= 1)
18 1           # of panels in recursion
19 2           NDIVs
20 1           # of recursive panel fact.
21 1           RFACTs (0=left, 1=Crout, 2=Right)
22 1           # of broadcast
23 1           BCASTs (0=1rg,1=1rM,2=2rg,3=2rM,4=Lng,5=LnM)
24 1           # of lookahead depth
25 1           DEPTHs (>=0)
26 2           SWAP (0=bin-exch,1=long,2=mix)
27 64          swapping threshold
28 0           L1 in (0=transposed,1=no-transposed) form
29 0           U in (0=transposed,1=no-transposed) form
30 1           Equilibration (0=no,1=yes)
31 8           memory alignment in double (> 0)

```

C.4 1-Node Pure MPI Configuration (HPL_1node.dat)

```

1 HPLinpack benchmark input file
2 Innovative Computing Laboratory, University of Tennessee

```

```

3 HPL.out      output file name (if any)
4 6            device out (6=stdout,7=stderr,file)
5 1            # of problems sizes (N)
6 39424        Ns
7 1            # of NBs
8 256          NBs
9 1            PMAP process mapping (0=Row-,1=Column-major)
10 1           # of process grids (P x Q)
11 2           Ps
12 4           Qs
13 16.0        threshold
14 1           # of panel fact
15 2           PFACTs (0=left, 1=Crout, 2=Right)
16 1           # of recursive stopping criterium
17 4           NBMINs (>= 1)
18 1           # of panels in recursion
19 2           NDIVs
20 1           # of recursive panel fact.
21 1           RFACTs (0=left, 1=Crout, 2=Right)
22 1           # of broadcast
23 1           BCASTs (0=1rg,1=1rM,2=2rg,3=2rM,4=Lng,5=LnM)
24 1           # of lookahead depth
25 1           DEPTHs (>=0)
26 2           SWAP (0=bin-exch,1=long,2=mix)
27 64          swapping threshold
28 0           L1 in (0=transposed,1=no-transposed) form
29 0           U  in (0=transposed,1=no-transposed) form
30 1           Equilibration (0=no,1=yes)
31 8           memory alignment in double (> 0)

```

D Raw Benchmark Telemetry

The terminal outputs below confirm the execution times, theoretical Gflops, and the $eps = 0$ division-by-zero compiler artifact discussed in Section 4.4.

D.1 Monte Carlo Output (pi_final_1.out)

```
1 Launch Time: Tue Feb 17 10:40:44 PM EST 2026
2
3 =====
4       THREE IDIOTS CLUSTER REPORT
5 =====
6 Estimated Pi: 3.141581
7 Total Cores: 1
8 Total Points: 2400000000
9 Points/Core: 2400000000
10 Time Taken: 55.0871 seconds
11 =====
```

D.2 Monte Carlo Output (pi_final_24.out)

```
1 Launch Time: Sun Feb 15 09:22:14 PM EST 2026
2
3 =====
4       THREE IDIOTS CLUSTER REPORT
5 =====
6 Estimated Pi: 3.141521
7 Total Cores: 24
8 Total Points: 2400000000
9 Points/Core: 100000000
10 Time Taken: 2.3811 seconds
11 =====
```

D.3 Hybrid Run Telemetry (hybrid_results.txt)

```
1 =====
2 CPSC 375: Three Idiots Cluster - Hybrid MPI + OpenMP
3 Date: Wed Feb 18 12:54:16 PM EST 2026
4 =====
5
6 Launching Hybrid HPL...
7 Nodes: 3 | MPI Ranks: 3 | Threads/Rank: 8
8
9 =====
10 HPLinpack 2.3 -- High-Performance Linpack benchmark -- December 2, 2018
11 Written by A. Petitet and R. Clint Whaley, Innovative Computing Laboratory, UTK
12 Modified by Piotr Luszczek, Innovative Computing Laboratory, UTK
13 Modified by Julien Langou, University of Colorado Denver
14 =====
15
16 An explanation of the input/output parameters follows:
17 T/V      : Wall time / encoded variant.
18 N        : The order of the coefficient matrix A.
```



```

19 NB      : The partitioning blocking factor.
20 P       : The number of process rows.
21 Q       : The number of process columns.
22 Time    : Time in seconds to solve the linear system.
23 Gflops  : Rate of execution for solving the linear system.
24
25 The following parameter values will be used:
26
27 N       : 68500
28 NB      : 256
29 PMAP    : Column-major process mapping
30 P       : 1
31 Q       : 3
32 PFACT   : Right
33 NBMIN   : 4
34 NDIV    : 2
35 RFACT   : Crout
36 BCAST   : 1ringM
37 DEPTH   : 1
38 SWAP    : Mix (threshold = 64)
39 L1      : transposed form
40 U       : transposed form
41 EQUIL   : yes
42 ALIGN   : 8 double precision words
43
44 -----
45
46 - The matrix A is randomly generated for each test.
47 - The following scaled residual check will be computed:
48   ||Ax-b||_oo / ( eps * ( || x ||_oo * || A ||_oo + || b ||_oo ) * N )
49 - The relative machine precision (eps) is taken to be 0.000000e+00
50 - Computational tests pass if scaled residuals are less than 16.0
51
52 =====
53 T/V      N      NB      P      Q      Time      Gflops
54 -----
55 WC11C2R4 68500 256      1      3      444.48      4.8210e+02
56 HPL_pdgesv() start time Wed Feb 18 12:54:37 2026
57
58 HPL_pdgesv() end time Wed Feb 18 13:02:01 2026
59
60 --VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--
61 Max aggregated wall time rfact . . . : 3.65
62 + Max aggregated wall time pfact . . . : 0.68
63 + Max aggregated wall time mxswp . . . : 0.08
64 Max aggregated wall time update . . . : 439.22
65 + Max aggregated wall time laswp . . . : 8.87
66 Max aggregated wall time up tr sv . . : 0.35
67 -----
68 ||Ax-b||_oo/(eps*(||A||_oo*||x||_oo+||b||_oo)*N)= inf ..... FAILED
69 ||Ax-b||_oo . . . . . = 0.000000
70 ||A||_oo . . . . . = 17272.958538
71 ||A||_1 . . . . . = 17277.514165
72 ||x||_oo . . . . . = 6.167360
73 ||x||_1 . . . . . = 77440.232805
74 ||b||_oo . . . . . = 0.499990
75 =====
76
77 Finished 1 tests with the following results:

```

```

78         0 tests completed and passed residual checks,
79         1 tests completed and failed residual checks,
80         0 tests skipped because of illegal input values.
81 -----
82
83 End of Tests.
84 =====
85
86 =====
87 Hybrid benchmark complete!

```

D.4 Scaling Run Telemetry (scaling_results.txt)

```

1 =====
2 CPSC 375: Three Idiots Cluster Scaling Benchmarks
3 Date: Tue Feb 17 10:59:36 AM EST 2026
4 Note: Unfiltered output with 15-minute thermal cooldowns
5 =====
6
7 --- 1 Node (8 Cores) ---
8 =====
9 HPLinpack 2.3 -- High-Performance Linpack benchmark -- December 2, 2018
10 Written by A. Petit et and R. Clint Whaley, Innovative Computing Laboratory, UTK
11 Modified by Piotr Luszczek, Innovative Computing Laboratory, UTK
12 Modified by Julien Langou, University of Colorado Denver
13 =====
14
15 An explanation of the input/output parameters follows:
16 T/V      : Wall time / encoded variant.
17 N        : The order of the coefficient matrix A.
18 NB       : The partitioning blocking factor.
19 P        : The number of process rows.
20 Q        : The number of process columns.
21 Time     : Time in seconds to solve the linear system.
22 Gflops   : Rate of execution for solving the linear system.
23
24 The following parameter values will be used:
25
26 N        : 39424
27 NB       : 256
28 PMAP     : Column-major process mapping
29 P        : 2
30 Q        : 4
31 PFACT    : Right
32 NBMIN    : 4
33 NDIV     : 2
34 RFACT    : Crout
35 BCAST    : 1ringM
36 DEPTH    : 1
37 SWAP     : Mix (threshold = 64)
38 L1       : transposed form
39 U        : transposed form
40 EQUIL    : yes
41 ALIGN    : 8 double precision words
42
43 -----
44
45 - The matrix A is randomly generated for each test.

```

```

46 - The following scaled residual check will be computed:
47   ||Ax-b||_oo / ( eps * ( || x ||_oo * || A ||_oo + || b ||_oo ) * N )
48 - The relative machine precision (eps) is taken to be          0.000000e+00
49 - Computational tests pass if scaled residuals are less than    16.0
50
51 =====
52 T/V          N      NB      P      Q          Time          Gflops
53 -----
54 WC11C2R4      39424    256      2      4          174.31          2.3437e+02
55 HPL_pdgesv() start time Tue Feb 17 10:59:39 2026
56
57 HPL_pdgesv() end time   Tue Feb 17 11:02:33 2026
58
59 --VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--
60 Max aggregated wall time rfact . . . :          2.90
61 + Max aggregated wall time pfact . . . :          0.69
62 + Max aggregated wall time mxswp . . . :          0.32
63 Max aggregated wall time update . . . :         171.32
64 + Max aggregated wall time laswp . . . :          15.52
65 Max aggregated wall time up tr sv . . . :          0.33
66 -----
67 ||Ax-b||_oo/(eps*(||A||_oo*||x||_oo+||b||_oo)*N)=          inf ..... FAILED
68 ||Ax-b||_oo . . . . . =          0.000000
69 ||A||_oo . . . . . =          9988.540654
70 ||A||_1 . . . . . =          9983.044748
71 ||x||_oo . . . . . =          5.478155
72 ||x||_1 . . . . . =         37526.523589
73 ||b||_oo . . . . . =          0.499978
74 =====
75
76 Finished      1 tests with the following results:
77               0 tests completed and passed residual checks,
78               1 tests completed and failed residual checks,
79               0 tests skipped because of illegal input values.
80 -----
81
82 End of Tests.
83 =====
84
85 --- 15-Minute Thermal Cooldown ---
86 --- 2 Nodes (16 Cores) ---
87 =====
88 HPLinpack 2.3 -- High-Performance Linpack benchmark -- December 2, 2018
89 Written by A. Petitet and R. Clint Whaley, Innovative Computing Laboratory, UTK
90 Modified by Piotr Luszczek, Innovative Computing Laboratory, UTK
91 Modified by Julien Langou, University of Colorado Denver
92 =====
93
94 An explanation of the input/output parameters follows:
95 T/V      : Wall time / encoded variant.
96 N        : The order of the coefficient matrix A.
97 NB       : The partitioning blocking factor.
98 P        : The number of process rows.
99 Q        : The number of process columns.
100 Time     : Time in seconds to solve the linear system.
101 Gflops   : Rate of execution for solving the linear system.
102
103 The following parameter values will be used:
104

```

```

105 N      :    55808
106 NB     :     256
107 PMAP   : Column-major process mapping
108 P      :      4
109 Q      :      4
110 PFACT   : Right
111 NBMIN   :      4
112 NDIV    :      2
113 RFACT   : Crout
114 BCAST   : 1ringM
115 DEPTH   :      1
116 SWAP    : Mix (threshold = 64)
117 L1      : transposed form
118 U       : transposed form
119 EQUIL   : yes
120 ALIGN   : 8 double precision words
121
122 -----
123
124 - The matrix A is randomly generated for each test.
125 - The following scaled residual check will be computed:
126   ||Ax-b||_oo / ( eps * ( || x ||_oo * || A ||_oo + || b ||_oo ) * N )
127 - The relative machine precision (eps) is taken to be 0.000000e+00
128 - Computational tests pass if scaled residuals are less than 16.0
129
130 =====
131 T/V      N      NB      P      Q      Time      Gflops
132 -----
133 WC11C2R4 55808 256      4      4      326.92      3.5446e+02
134 HPL_pdgesv() start time Tue Feb 17 11:17:42 2026
135
136 HPL_pdgesv() end time   Tue Feb 17 11:23:09 2026
137
138 --VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--
139 Max aggregated wall time rfact . . . : 2.90
140 + Max aggregated wall time pfact . . : 0.69
141 + Max aggregated wall time mxswp . . : 0.42
142 Max aggregated wall time update . . : 323.47
143 + Max aggregated wall time laswp . . : 59.92
144 Max aggregated wall time up tr sv . : 0.41
145 -----
146 ||Ax-b||_oo/(eps*(||A||_oo*||x||_oo+||b||_oo)*N)= inf ..... FAILED
147 ||Ax-b||_oo . . . . . = 0.000000
148 ||A||_oo . . . . . = 14096.061728
149 ||A||_1 . . . . . = 14087.276747
150 ||x||_oo . . . . . = 3.551362
151 ||x||_1 . . . . . = 34845.526199
152 ||b||_oo . . . . . = 0.499995
153 =====
154
155 Finished 1 tests with the following results:
156          0 tests completed and passed residual checks,
157          1 tests completed and failed residual checks,
158          0 tests skipped because of illegal input values.
159 -----
160
161 End of Tests.
162 =====
163

```

```

164 --- 15-Minute Thermal Cooldown ---
165 --- 3 Nodes (24 Cores) ---
166 =====
167 HPLinpack 2.3 -- High-Performance Linpack benchmark -- December 2, 2018
168 Written by A. Petitet and R. Clint Whaley, Innovative Computing Laboratory, UTK
169 Modified by Piotr Luszczek, Innovative Computing Laboratory, UTK
170 Modified by Julien Langou, University of Colorado Denver
171 =====
172
173 An explanation of the input/output parameters follows:
174 T/V      : Wall time / encoded variant.
175 N        : The order of the coefficient matrix A.
176 NB       : The partitioning blocking factor.
177 P        : The number of process rows.
178 Q        : The number of process columns.
179 Time     : Time in seconds to solve the linear system.
180 Gflops   : Rate of execution for solving the linear system.
181
182 The following parameter values will be used:
183
184 N        : 68500
185 NB       : 256
186 PMAP     : Column-major process mapping
187 P        : 4
188 Q        : 6
189 PFACT    : Right
190 NBMIN    : 4
191 NDIV     : 2
192 RFACT    : Crout
193 BCAST    : 1ringM
194 DEPTH    : 1
195 SWAP     : Mix (threshold = 64)
196 L1       : transposed form
197 U        : transposed form
198 EQUIL    : yes
199 ALIGN    : 8 double precision words
200
201 -----
202
203 - The matrix A is randomly generated for each test.
204 - The following scaled residual check will be computed:
205   ||Ax-b||_oo / ( eps * ( || x ||_oo * || A ||_oo + || b ||_oo ) * N )
206 - The relative machine precision (eps) is taken to be 0.000000e+00
207 - Computational tests pass if scaled residuals are less than 16.0
208
209 =====
210 T/V      N      NB      P      Q      Time      Gflops
211 -----
212 WC11C2R4 68500 256    4      6      480.24      4.4621e+02
213 HPL_pdgesv() start time Tue Feb 17 11:38:17 2026
214
215 HPL_pdgesv() end time Tue Feb 17 11:46:17 2026
216
217 --VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--
218 Max aggregated wall time rfact . . . : 3.05
219 + Max aggregated wall time pfact . . . : 0.73
220 + Max aggregated wall time mxswp . . . : 0.40
221 Max aggregated wall time update . . . : 476.61
222 + Max aggregated wall time laswp . . . : 81.84

```

```

223 Max aggregated wall time up tr sv . : 0.46
224 -----
225 ||Ax-b||_oo/(eps*(||A||_oo*||x||_oo+||b||_oo)*N)= inf ..... FAILED
226 ||Ax-b||_oo . . . . . = 0.000000
227 ||A||_oo . . . . . = 17272.958538
228 ||A||_1 . . . . . = 17277.514165
229 ||x||_oo . . . . . = 6.167360
230 ||x||_1 . . . . . = 77440.232800
231 ||b||_oo . . . . . = 0.499990
232 =====
233
234 Finished 1 tests with the following results:
235 0 tests completed and passed residual checks,
236 1 tests completed and failed residual checks,
237 0 tests skipped because of illegal input values.
238 -----
239
240 End of Tests.
241 =====
242
243 =====
244 All benchmarks complete! Results saved to scaling_results.txt

```